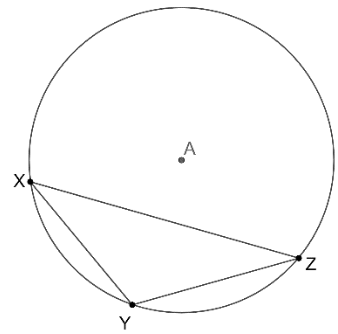
Geometry
Unit 11
Lesson 3 Homework

Name Answer Key
Date____________________
Period___________________

Circle A is shown, where points $X, Y,$ and Z are on circle A and create ΔXYZ. $m\widehat{XZ} = 240°$, $m\angle ZXY = 20°$, and $m\angle XYZ = (4x - 28)°$.



1. Find the value of x.

$$\frac{2(4x-28) = 240}{2}$$
$$4x - 28 = 120$$
$$+28 = +28$$
$$\frac{4x}{4} = \frac{148}{4}$$
$$x = 37$$

2. Find the $m\angle XYZ$.

$$\frac{240}{2} = 120°$$

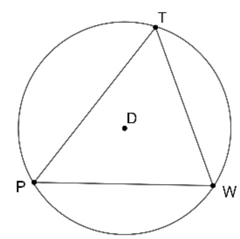
3. Find the $m\widehat{YZ}$.

$2(20) = 40°$

4. Find $m\angle YZX$.

$20 + 120 + m\angle YZX = 180$
$140 + m\angle YZX = 180$
$m\angle YZX = 40°$

Circle D is shown, where points $P, T,$ and W are on circle D and create ΔPTW. $m\angle PTW = 54°$ and $m\angle TWP = 60°$.



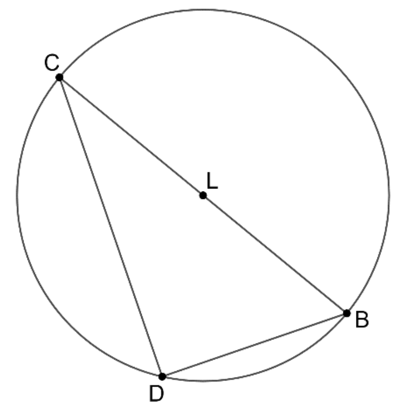
5. Find $m\widehat{PT}$.

$2(60) = 120°$

6. Find $m\widehat{TW}$.

$54 + 60 + m\angle P = 180$
$114 + m\angle P = 180$
$m\angle P = 66°$
$2(66) = 132°$

Circle L has points $B, C,$ and D on circle L and diameter $\overline{BC}$. ΔBCD is inscribed in the circle. $m\widehat{CD} = 100°$, $m\angle DCB = 40°$, and $m\angle DBC = (2x + 12)°$.



7. What is the $m\angle CDB$?

$90°$

8. What is the value of x?

$2x + 12 = 50$

$-12 = -12$

$\frac{2x}{2} = \frac{38}{2}$

$x = 19$

9. Determine $m\angle DBC$.

$\frac{100}{2} = 50°$

10. Determine $m\widehat{DBC}$?

$180 + 80 = 260°$